Continuity and Differentiability Cheat Sheet

A comprehensive summary

1. Functions, Domains, and Limits

1.1. Basic Definitions

- **Function ($f : A \to B$):** A rule assigning every element of subset $A \subseteq \mathbb{R}$ to a unique element of $B \subseteq \mathbb{R}$.
- **Range:** Subset of B containing images of all elements of A. $f(A) = \{f(x) \mid x \in A\}$.
- **Domain of Definition (DOD):** Subset $S \subseteq \mathbb{R}$ where a real-valued function is defined.
- **One-One (Injective):** $f(x_1) = f(x_2) \implies x_1 = x_2$.
- **Onto (Surjective):** Range equals Co-domain.

1.2. Well-Known Functions

- **Constant:** $f(x) = a$
- **Square Root:** $f(x) = \sqrt{x}$ (DOD: $x \geq 0$)
- **Absolute Value:** $f(x) = |x|$
- **Signum:** $f(x) = 1$ if $x > 0$, 0 if $x = 0$, -1 if $x < 0$
- **Greatest Integer:** $f(x) = [x]$
- **Fractional Part:** $f(x) = \{x\} = x - [x]$
- **Logarithmic:** $f(x) = \log_a x$

1.3. Limit of a Function

$L \in \mathbb{R}$ is a limit of f as $x \to a$ if for any $\varepsilon > 0$, $\exists \delta > 0$ such that $0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$.

- Limit exists at $x = a$ *iff* Right Hand Limit (RHL) = Left Hand Limit (LHL).

2. Continuity

f is continuous at $x = a$ if $\lim_{x \to a} f(x) = f(a)$.

2.1. Types of Discontinuity

- **Removable Discontinuity:** Limit exists but $\neq f(a)$. Can be removed by redefining $f(a)$.
- **First Kind (Jump):** RHL and LHL both exist but are unequal.
- **Second Kind:** At least one of RHL or LHL fails to exist.
- **Mixed Discontinuity:** Second kind on one side, first kind on the other.

2.2. Important Theorems

- **Boundedness Theorem:** If f is continuous on closed interval I, it is bounded on I.
- **Location of Roots:** If f is continuous on $[a,b]$ and $f(a)f(b) < 0$, $\exists c \in (a,b)$ such that $f(c) = 0$.
- **Intermediate Value Theorem:** If f is continuous on $[a,b]$ and $f(a) \neq f(b)$, f assumes every value between $f(a)$ and $f(b)$.
- **Fixed Point Theorem:** If $f : [a,b] \to [a,b]$ is continuous, $\exists c \in [a,b]$ such that $f(c) = c$.

3. Uniform Continuity

f is uniformly continuous if for given $\varepsilon > 0$, $\exists \delta > 0$ (depending **only** on ε) such that $|f(x_1) - f(x_2)| < \varepsilon$ whenever $|x_1 - x_2| < \delta$.

- Continuous on closed bounded interval $\implies$ uniformly continuous.
- Continuous on (a,b) $\implies$ uniformly continuous iff $\lim_{x\to a^+} f(x)$ and $\lim_{x\to b^-} f(x)$ exist.
- Uniformly continuous function may not be bounded.

4. Differentiability

f is derivable at c if $f'(c) = \lim_{x\to c} \frac{f(x)-f(c)}{x-c}$ exists (LHD = RHD).

- Every differentiable function is continuous. Converse is not true.
- **Convex Function:** $f''(x) \geq 0$. **Concave:** $f''(x) \leq 0$.
- **Point of Inflexion:** $f''(a) = 0$ (and changes sign).

4.1. Important Theorems & Applications

- **Darboux's Theorem:** If f is derivable on $[a,b]$ and $f'(a) < k < f'(b)$, then $\exists c \in (a,b)$ such that $f'(c) = k$.
- **Rolle's Theorem:** If f is continuous on $[a,b]$, derivable on (a,b), and $f(a) = f(b)$, then $\exists c \in (a,b)$ such that $f'(c) = 0$.
- **Lagrange's Mean Value Theorem (LMVT):** $\exists c \in (a,b)$ such that $\frac{f(b)-f(a)}{b-a} = f'(c)$.
- **Cauchy's Mean Value Theorem (CMVT):** $\exists c \in (a,b)$ such that $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$.
- **Taylor's Theorem (Lagrange Remainder):** $R_n = \frac{h^n}{n!} f^{(n)}(a+\theta h)$.

5. Useful Tricks

- **Trick 1:** If $f : \mathbb{R} \to \mathbb{R}$ is continuous and one-one, then f is strictly monotonic.
- **Trick 2:** If $f : \mathbb{R} \to \mathbb{R}$ is an increasing function, the set of points of discontinuity is at most countable.